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MUTATION IN BRIEF

Updated Comprehensive Phylogenetic Tree of Global Human Mitochondrial DNA Variation

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Human mitochondrial DNA is widely used as tool in many fields including evolutionary anthropology and population history, medical genetics, genetic genealogy, and forensic science. Many applications require detailed knowledge about the phylogenetic relationship of mtDNA variants. Although the phylogenetic resolution of global human mtDNA diversity has greatly improved as a result of increasing sequencing efforts of complete mtDNA genomes, an updated overall mtDNA tree is currently not available. In order to facilitate a better use of known mtDNA variation, we have constructed an updated comprehensive phylogeny of global human mtDNA variation, based on both coding- and control region mutations. This complete mtDNA tree includes previously published as well as newly identified haplogroups, is easily navigable, will be continuously and regularly updated in the future, and is online available at http://www.phylotree.org. © 2008 Wiley-Liss, Inc.

KEY WORDS: mitochondrial DNA, phylogenetic tree, haplogroup, mtDNA variation, human evolution

INTRODUCTION

Uniparentally inherited portions of the human genome have important applications in a wide range of fields including evolutionary anthropology and population history (Jobling and Tyler-Smith, 2003; Torroni et al., 2006; Underhill and Kivisild, 2007), medical genetics (Krausz et al., 2004; Taylor and Turnbull, 2005), genetic genealogy (Blansit, 2006; Johnston and Thomas, 2003; Shriver and Kittles, 2004) as well as forensic science (Kayser, 2007). Recently, an updated haplogroup tree of the patrilineally inherited, non-recombining part of the human Y chromosome (NRY) was published (Karafet et al., 2008). Here, we present an updated phylogenetic tree of its matrilineal counterpart, the mitochondrial (mt)DNA.

PROPERTIES OF mtDNA

MtDNA represents the small fraction of our genome that does not reside in the cell's nucleus, but is contained in the energy-producing mitochondria. It is a circular, double-stranded molecule with a length of nearly 16.6 kilobase pairs that encodes proteins required for oxidative phosphorylation. Special features of mtDNA include matrilineal inheritance, lack of recombination, high copy number, and a higher evolutionary turn over rate as compared to nuclear DNA. It is assumed that all mtDNA types in the human gene pool can ultimately be traced

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back to a common matrilineal ancestor that lived approximately 200,000 years ago in Africa (Behar et al., 2008b; Macaulay et al., 2005; Mishmar et al., 2003). MtDNA sequence variation thus evolved as a result of the sequential accumulation of mutations along maternally inherited lineages, which can be represented in a tree reflecting the phylogenetic relationships of known mtDNA variants. The importance of a detailed global mtDNA phylogeny has been addressed by several authors (Bandelt et al., 2005, 2006; Kivisild et al., 2006a; Kong et al., 2006; Richards, 2004; Salas et al., 2007; Torroni et al., 2006). Recently, a global mtDNA phylogeny based on 2959 coding region sequences was published as part of the MITOMAP database (Ruiz-Pesini et al., 2007). However, this tree does not include the commonly used control region mutations and does not contain a detailed haplogroup nomenclature. Moreover, the MITOMAP tree must be considered outdated by now since many complete mtDNA sequences that have considerably improved the resolution to the mtDNA phylogeny became available only after its publication. For instance, Behar et al. (2008b) recently reported 309 complete mtDNA sequences that (combined with 315 previously published sequences) greatly increased the resolution of haplogroup L. At present (August 27, 2008), 4198 complete mtDNA genome sequences and additional 945 separate coding region sequences are available via the NCBI GenBank database (http://www.ncbi.nlm.nih.gov/GenBank). In order to facilitate a better use of the known mtDNA variation, we generated an updated phylogenetic tree of global human mtDNA variation considering all currently available sequence data.

CODING VERSUS CONTROL REGION

By convention, the nucleotide positions of the human mtDNA genome are numbered from 1 to 16569 according to the revised Cambridge Reference Sequence (rCRS) (Anderson et al., 1981; Andrews et al., 1999). Detected polymorphisms (single nucleotide polymorphisms [SNPs] and insertions/deletion polymorphisms [indels]) are then scored as differences ("mutations") to the rCRS. The mutation rate is not equal across the entire mtDNA molecule; the overall mutation rate in the 1.1 kb non-coding control region (bases 16024-576) is about 10 times higher than that of the 15.5 kb coding region (bases 577-16023) (Howell et al., 2007; Pakendorf and Stoneking, 2005). This can be explained by purifying selection filtering out from the population the often deleterious functional mutations. In contrast, the control region, which is important for mtDNA replication, is non-coding and thus accumulates more mutations. Moreover, local differences in the mutation rate among nucleotide positions within the control region exist as well (Hasegawa et al., 1993; Meyer et al., 1999). Some individual sites act as mutational hot spots (e.g. positions 146, 150, 152, 195, 16189, 16311, 16362, 16519), whereas others appear rather stable (e.g. positions 477, 493, 16108, 16219). Also within the coding region there are sites that are more prone to mutation than others (e.g. positions 709, 1719, 3010, 5460, 10398, 11914, 13105, 13708, 15884), causing homoplasy (recurrence) in the phylogeny. Although several authors have addressed the issue of site-specific (hyper)mutability (Galtier et al., 2006; Howell et al., 2007; Ingman and Gyllensten, 2007; Kivisild et al., 2006b; Malyarchuk et al., 2002), the exact mechanism accounting for it remains to be elucidated. Due to the higher overall mutation rate, the control region is relatively enriched in sequence variation, and therefore researchers typically sequence (part of) this region for various applications. However, at least some haplogroups can not be confidently assigned based on control region data only (e.g. H4, V), whereas for others (e.g. J1b1, K1a9) the accurate inference solely from control region sequence data is feasible. Therefore it is relevant to confirm predicted haplogroups based on control region sequences by use of informative SNPs from the coding region and/or to perform initial mtDNA haplogroup assignment using coding region information either solely or combined with control region data. However, ascertaining informative coding region SNPs strongly depends on the availability and quality of phylogenetic information, hence requires an updated phylogenetic global mtDNA tree.

HAPLOGROUP NOMENCLATURE

The first mtDNA haplogroups, discovered in Native Americans, were baptized A, B, C, and D (Torroni et al., 1993). Subsequently detected haplogroups were designated using other letters of the alphabet. By now, all letters of the alphabet, except O (although once proposed), have been used (Figure 1). Simple rules for mtDNA haplogroup nomenclature were proposed by Richards et al. (1998) (see also Kivisild et al., 2006a); but unfortunately, the mtDNA haplogroup nomenclature is not always used consistently. Several examples can be found where different authors coined the same name for different haplogroups: e.g. "M12" in Kong et al. (2006) versus Metspalu et al. (2006), Tanaka et al. (2004); "R1" in Malyarchuk et al. (2008a), Palanichamy et al. (2004) versus Metspalu et al. (2006); "S" in Hudjashov et al. (2007), Palanichamy et al. (2004) versus Starikovskaya et al.

(2005). Others used different names for the same haplogroup: “N12” in Hudjashov et al. (2007) versus “O” in Palanichamy et al. (2004), Pierson et al. (2006); M1 sublineages in Olivieri et al. (2006) versus Gonzalez et al. (2007); “C4” in Achilli et al. (2008), Kong et al. (2006) versus “C2” in Starikovskaya et al. (2005), Volodko et al. (2008). All these examples illustrate the need for an overall mtDNA phylogeny with universal nomenclature but also their consistent use by the mtDNA community.

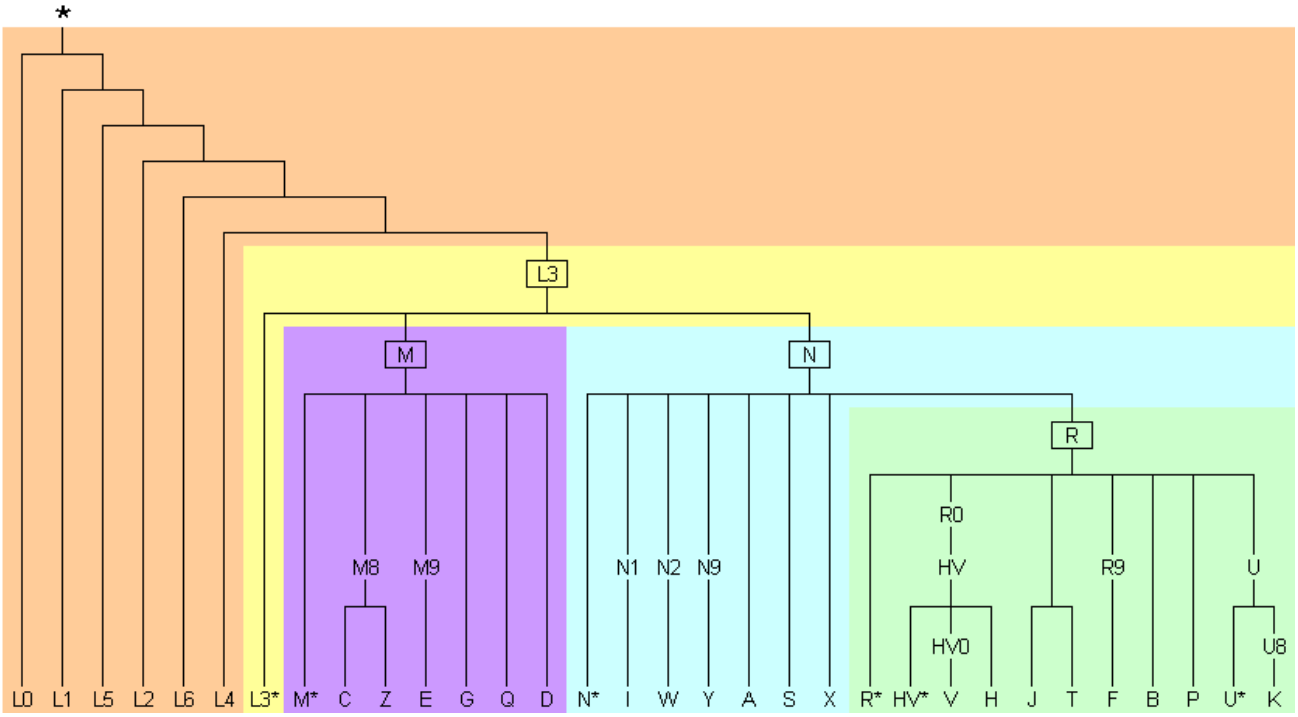


Figure 1. Simplified mtDNA phylogeny illustrating the use of alphabetical letters for haplogroup designation. All letters of the alphabet, except O (Palanichamy et al. [2004] proposed this label for a haplogroup that was later relabeled N12 by Hudjashov et al. [2007]), have been used so far. The root of the tree is indicated by a star representing the most recent common matrilineal ancestor of all humans. The L haplogroups are the most deep-rooting lineages and are African specific indicating the African origin of modern humans as well as the out-of-Africa exodus as also found based on other genetic as well as fossil data. Haplogroup L3 gave rise to macrohaplogroups M, N and R (the latter itself a subclade of N), which encompass all variation observed outside Africa. Nomenclature evolved in such a way that letters C, D, E, G, Q, and Z designate lineages belonging to M; letters A, I, S, W, X, and Y lineages within N; and B, F, HV, H, J, K, P, T, U, and V lineages within R. Haplogroup symbols followed by a star represent all other descendant lineages (besides the ones shown) of a particular clade, for which no unique alphabetical letters were reserved; e.g. N* stands for N5, N12, N13, N14, N21 and N22. Note that this tree should not be seen as a summary of global mtDNA variation (for this see the tree in the supporting information or at http://www.phylotree.org); it is merely meant to show the topology of alphabetically named haplogroups.

UPDATED COMPLETE mtDNA TREE

We have consulted 55 publications that have recently described mtDNA variation based on complete mtDNA sequences to construct an updated and complete (as per August 27, 2008) global mtDNA phylogenetic tree (the tree’s references are listed in the main reference list of this article and are as separate list also available at http://www.phylotree.org). In cases of conflicting information from different papers concerning the same haplogroup, we have drawn the phylogeny ourselves by comparing the maximum number of available mtDNA sequences of that particular haplogroup, in order to get the full up-to-date picture. In addition, we made use of other recently published complete mtDNA sequences (FamilyTreeDNA; Gasparre et al., 2007, Hartmann et al., 2008; Just et al., 2008) that were not previously incorporated in phylogenetic schemes (but were available at GenBank). In some cases this allowed us to further increase the phylogenetic resolution by adding novel subbranches to the tree (i.e. haplogroups L3b1b1, L3d5, Q3a1, M33c, M52, HV0c, H2a2b and H2a2b1). We used

as a general criterion for the whole tree that a relatively stable (set of) mutation(s) must be shared by at least three complete sequences before assigning it the haplogroup status. We deviated from this rule in the following cases: some authors have defined haplogroups based on hypervariable positions (e.g. M4, X1) or on incomplete mtDNA sequence data (e.g. M21d, R22) and in those cases we decided to follow the established nomenclature; when less than three complete sequences added a relatively deep-rooting branch to the tree we also found it justified to assign it a haplogroup name (e.g. L1c1a1b, M52). For clarity, all branches that are insufficiently supported by complete mtDNA sequence data were given a preliminary status by showing their defining mutations in italic. The final mutation motifs of these haplogroups will be further refined when additional data accumulates. In total, our tree encompasses 1267 branching points, of which 657 are terminal branches. In contrast to what has become common practice for drawing mtDNA phylogenetic trees so far, we decided to use horizontal instead of vertical lines representing the branches of the tree. Not only was such representation used for most NRY trees, including the most up-to-date one by Karafet et al. (2008), but the horizontal representation results in a more compact tree, which we find easier to follow. For each branch its defining mutations were mentioned as well as the haplogroup name (newly proposed haplogroups were indicated in red). Coding region mutations were indicated in black and control region mutations were in blue, so that it becomes immediately clear which information can be derived from which part of the mtDNA. Position numbers were relative to the rCRS and the scoring method followed the guidelines outlined by Bandelt and Parson (2008). At the tip of each branch we provided references to the papers that were consulted for the particular haplogroups. The references given are not necessarily the papers in which the haplogroup was first described, but rather the papers in which the more detailed phylogenies were provided. For each terminal branch we also provided one or two NCBI accession numbers (if available) of complete mtDNA sequences belonging to the respective branch. See Figure 2 for a snapshot view of the tree we constructed. The complete updated tree is available as Supporting Information to this paper and additionally freely available at http://www.phylotree.org. We plan to regularly update this tree and keep the further updated versions available via this website. The tree can be viewed with the most common internet browsers. A convenient way to look up a particular haplogroup or a particular mutation is to use the "Find" command of the web browsers. Internet Explorer has the advantage that one can select the option "Match whole word only". This is recommended to avoid unspecific hits in cases where a searched position number also occurs as part of other position numbers (e.g. to avoid finding "11719" when searching for "1719").

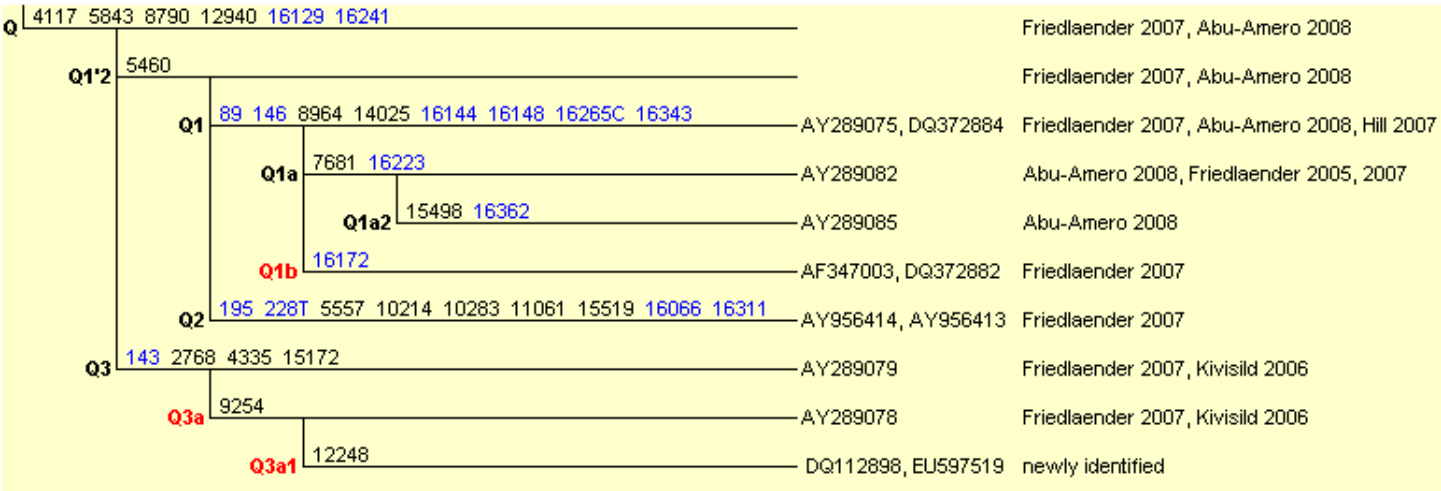


Figure 2. Snapshot of the haplogroup Q branch from the updated global mtDNA tree available as supplement of this paper as well as at http://www.phylotree.org. Coding region mutations were labeled in black, control region mutations were in blue. Mutations are transitions unless a specific base change was specified. Haplogroup names were given at the left of each branch. Haplogroups were indicated in black when previously published and in red when newly proposed by us. At the tip of each branch we provided the references that were consulted for the particular haplogroups, as well as GenBank accession numbers of underlying complete mtDNA sequences. In the snapshot example shown here we proposed a new haplogroup Q3a1 defined by a transition at position 12248, based on the fact that this mutation was shared by the sequences with accession numbers DQ112898 and EU597519.

CONCLUSIONS

We have constructed an updated, high-resolution global mtDNA phylogenetic tree based on both coding and control region data, including universal haplogroup nomenclature, and providing references to consulted papers as well as to accession numbers of underlying NCBI GenBank sequences. This tree is publicly available at http://www.phylotree.org, allows easy navigation via search functions of internet browsers and will be updated at least every six months on the website. This comprehensive mtDNA tree is meant as a framework for evolutionary anthropologists, population geneticists, medical geneticists, genealogists, forensic geneticists and other scientists interested in the description and application of human mtDNA diversity.

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